

DATA EXTRACTION FLOW

1. **Extract UDP 443 Traffic.**
2. **Filter for Long Headers:** Check the first byte (0x80 or higher).
3. **Identify SNI:** Decrypt the Initial packet to label the "Provider" (Google/Facebook).
4. **Transition to Short Headers:** Once the connection switches to Short Headers (0x40), stop looking at the content (as it's now fully encrypted) and start looking at **Inter-Arrival Times (IAT)** and **Packet Lengths**.
5. **Compute Statistics:**
 - If 90% of packets are > 1200 bytes and the flow lasts > 30 seconds -> **Label: Streaming.**
 - If there is a burst of 50 packets and then silence -> **Label: Web.**

5. Generic UDP (The "Baseline")

To ensure your classifier is accurate, compare the above to:

- **DNS:** UDP 53, very small, one packet in/one packet out.
- **VoIP (non-QUIC):** Consistent packet size (~200 bytes) every 20ms exactly.

Types of Streams that QUIC Traffic Carries

- **Request/Response Bodies:** The actual bytes of the website, video, or file.
- **Header Blocks:** Compressed metadata (via QPACK) that describes the request or response.
- **Control Frames:** Protocol-level instructions (SETTINGS, CANCEL_PUSH, MAX_DATA) that coordinate the two endpoints.
- **Push Promises:** Information from the server about resources it intends to "push" to the client.

PROOF OF CONCEPT

CAPTURE PACKETS

Generic UDP

QUIC - Streaming

QUIC - Web

FEATURE EXTRACTION

Extract Shape, Size and Time (SST) and other details, and store in a CSV for further analysis.

Add Label Column, for training purpose.

Library Used: **nfstream**

NFStream does the heavy lifting for the ML model:

- **Flow Grouping:** It finds all packets belonging to one connection.
- **Statistical Calculation:** It looks at the **Mean Packet Size** (Size), the **IAT Standard Deviation** (Timing), and the **Byte Ratios** (Shape).
- **Labeling:** It adds a column called label, so the computer knows which numbers belong to "Streaming", "Web", and "Generic UDP."

label	detected_protocol	filesource	bidirectional	bidirectional	bidirectional	bidirectional_bidirectional_stddev_p	bidirectional	bidirectional	bidirectional
Streaming	QUIC	youtube_1.pcapng	8	557	54080	68.625	1.505940617307715	7722.857	142857142
Streaming	QUIC	youtube_1.pcapng	892	60749	53857	68.104391585	54546335619715	50.4455667	7890011
Streaming	QUIC	youtube_1.pcapng	30	2065	658459	68.8333333	2.5875475452347874	2260.31034	48275965
Streaming	QUIC	youtube_1.pcapng	18	5263	12935	292.38888	400.87025022610476	760.8823529411765	3062.83595876
Streaming	QUIC	youtube_1.pcapng	538	586264	15894	1089.71070	439.0092976660548	29.59775636312848	422.313822614
Streaming	QUIC:Google	youtube_1.pcapng	31	17132	71	552.64515	578.794180366212	2.366666666666663	4.8714634768
Streaming	QUIC	youtube_1.pcapng	7	1046	126	149.42857	140.10693195182162		21.3141337294844
Streaming	QUIC:Google	youtube_1.pcapng	29	17326	186	597.48427	577.5467566852362	6.642857	142857142
Streaming	QUIC:YouTube	youtube_1.pcapng	9142	9318047	52663	1019.2565	506.17760110347655	151.78105647	12.8369503040
Streaming	QUIC:YouTube	youtube_1.pcapng	3496	3750860	7331	1072.9004	466.90506807279877	2.09756754220315	68.2352653649
Streaming	QUIC	youtube_1.pcapng	18	6083	11794	373.94444	373.3872728890951	693.7647058823528	2789.35771122
Streaming	QUIC:YouTube	youtube_1.pcapng	744	789969	785	1061.7865	474.0940315669215	1.065627590849174	4.034825794602
Streaming	QUIC	youtube_1.pcapng	35	3468	4040	99.06571	52.6604018160525896	118.23252941176472	592.735439057
Streaming	QUIC:Google	youtube_1.pcapng	110	86011	3421	781.91815	585.65634993302184	31.38532110091742	209.649809815
Streaming	QUIC:YouTube	youtube_1.pcapng	57	45051	887	110.54395	588.33242151276014	15.8392857	14285715
Streaming	QUIC	youtube_1.pcapng	71	54225	11574	783.73235	591.4175696154954	38.342857	14285742
Streaming	QUIC:PlayStore	youtube_1.pcapng	28	12710	283	453.92857	527.4852940601619	10.481481481481483	23.29036566666
Streaming	QUIC	youtube_1.pcapng	4	280	27047	70	1.5470507339792515	9015.66666666666	15557.5748589
Streaming	QUIC:YouTube	youtube_1.pcapng	25	16575	359	663.564	208970816364	14.958333333333332	26.119265871
Web	QUIC:Google	web_1.pcapng	21	14256	97	678.8571	569.165642477259		4.857.77884677958
Web	QUIC:Google	web_1.pcapng	44	22283	473	506.43181	536.40293356254692	10.999999999999996	25.5063017423
Web	QUIC:Google	web_1.pcapng	27	15503	301	574.18151	568.4640362387536	11.576923079203078	21.6336276697
Web	QUIC:Google	web_1.pcapng	23	16150	185	702.17375	650.0650639003545	6.409009090909090	11.93088247911
Web	QUIC	web_1.pcapng	1714	1482037	53207	864.66565	565.2341865155114	31.060712200871728	466.281560835
Web	QUIC:Google	web_1.pcapng	23	14696	331	638.9565	575.05711000958545	15.045454545454545	30.1811269565
Web	QUIC:Google	web_1.pcapng	74	58477	105	790.22972	580.5578221481993	1.4383561643835616	3.68173532971
Web	QUIC:YouTube	web_1.pcapng	211	173394	990	821.7175	585.0041150875855	74.1428571	30.6277899534
Web	QUIC:YouTube	web_1.pcapng	20	31117	78	655.84968	583.3308418493169	4.105623157894736	7.07809403756
Web	QUIC:Google	web_1.pcapng	65	52321	226	804.93845	577.6110790908557	5.33125000000000013	13.89754888474
Web	QUIC:PlayStore	web_1.pcapng	27	16112	256	596.7407	564.7501558282615	9.846148148148148	21.4283780211
Web	QUIC:Google	web_1.pcapng	23	15970	276	694.34782	568.2837486435205	12.545454545454547	30.44077352233
Web	QUIC:Apple	web_1.pcapng	28	16054	85	573.3571	585.7458818423208	3.1481481481481484	7.68466857152
Web	QUIC	web_1.pcapng	35	24658	277	704.5126	554.8283980299713	6.147058823529413	14.01326720408
Generic_UDP	QUIC	generic_1.pcapng	1	1042	0	1042	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	1042	0	1042	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	1042	0	1042	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	52	0	52	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	1042	0	1042	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	52	0	52	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	1042	0	1042	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	1042	0	1042	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0
Generic_UDP	QUIC	generic_1.pcapng	1	53	0	53	0	0	0

TRAINING ML MODEL

Using the Random Forest Classifier model for training on this data.

```
(venv) rajanhasija~$python3 train_model.py
---- MODEL PERFORMANCE ----
Accuracy: 80.00%

--- Details ---
      precision    recall  f1-score   support

Generic_UDP      1.00      1.00      1.00         3
  Streaming      0.50      1.00      0.67         2
    Web          1.00      0.60      0.75         5

   accuracy      0.80      0.80      0.80        10
  macro avg      0.83      0.87      0.81        10
weighted avg      0.90      0.80      0.81        10

Success! Model saved!
```

TESTING THE MODEL

For testing the classification, two separate packets have been captured (web and youtube), completely new to the model. Here are the findings:

```
--- Analyzing File: youtube_1.pcapng ---
Total Flows Detected: 167
Streaming Detected: 47 times
Web Detected: 4
Generic UDP Detected: 116

--- Analyzing File: web_1.pcapng ---
Total Flows Detected: 207
Streaming Detected: 71 times
Web Detected: 8
Generic UDP Detected: 128
```

- **The Observation:** The Initial POC shows the model is highly sensitive to high-bandwidth flows. In web_1.pcapng, many flows were identified as Streaming.
- **The Technical Reason:** Modern web pages use CDN-hosted resources and high-res images that mimic the statistical footprint of a video chunk—specifically high Mean Packet Size and sustained Byte counts.
- **Plan:** To improve precision, the model can be retrained with a more diverse 'Web' dataset that includes modern, heavy websites to better distinguish them from actual continuous video streams.